



SafeGuard AI

AI-Powered Detection and Analysis of Toxic Communication Patterns

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Data Science MSIT - GenAI Advanced Module

The Problem

Isolated Analysis

Most toxicity detection systems evaluate single messages, missing persistent patterns that emerge over time.

No Educational Context

Systems label content as toxic without explaining what it means or providing educational resources.

Privacy Risks

Sending sensitive personal messages to external AI models raises serious GDPR and privacy concerns.

SafeGuard AI — Solution Overview

SafeGuard AI helps users identify and understand potentially harmful communication patterns in personal, educational, and workplace contexts.

- Multi-label toxic pattern classification
- Behavioral Pattern Timeline: tracks escalation over time
- Contextual risk assessment based on relationship history
- RAG-grounded educational explanations
- PII anonymisation before any AI call
- Response strategies for each situation
- Educational chat assistant

Core Principle

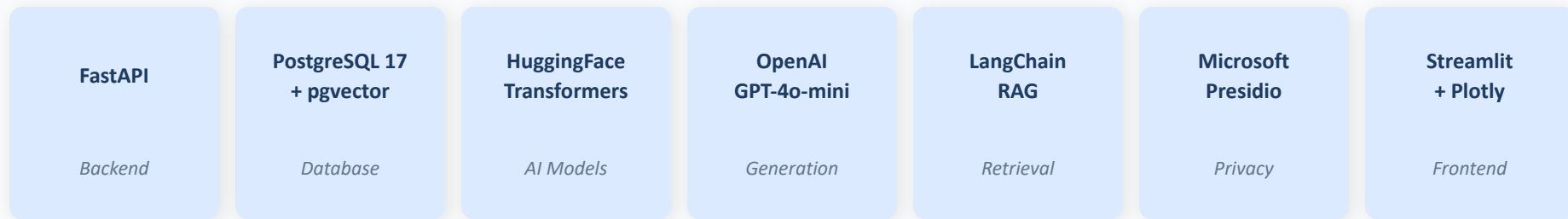
The system analyzes
linguistic patterns,
not people.

No diagnosis.
No guilt.
No legal advice.

System Architecture



Tech Stack



AI Pipeline — Two Distinct Models

Model 1 — HuggingFace

DistilBERT (fine-tuned)

Task: Binary gate — toxic / non-toxic

Training: civil_comments + daily_dialog + custom relational examples

Accuracy: 86.5% | F1: 84.7%

facebook/bart-large-mnli

Task: Zero-shot multi-label classification

Labels: 10 toxic communication categories
Risk score calculated across all 10 categories

Model 2 — OpenAI

GPT-4o-mini

Task: RAG-grounded explanation + response strategies

System prompt: full ethical policy embedded

RAG-constrained: answers based exclusively on retrieved educational documents

Output: Explanation + 3 response strategies
(Assertive / Neutral / De-escalation)

Retrieval-Augmented Generation (RAG)

RAG Pipeline

1. 13 educational documents
2. Chunking (500 tokens, 50 overlap)
3. OpenAI text-embedding-3-small
4. pgvector similarity search
5. Top 3 relevant chunks retrieved
6. GPT generates grounded response

90+ chunks indexed in pgvector
constrained RAG prevents hallucination

Knowledge Base Sources

thehotline.org - Coercive control, gaslighting, emotional abuse

loveisrespect.org - Relationship spectrum, healthy relationships

verywellmind.com - Manipulation tactics, boundaries, stalking

acas.org.uk - Workplace bullying

allvoices.co - Workplace mobbing

stopbullying.gov - Bullying, cyberbullying

skillsyouneed.com - Assertive communication

The Core Innovation - Behavioral Timeline

Why single messages are not enough

"Who were you with tonight?"

In isolation:

ambiguous, could be genuine curiosity

→ Message risk: None

With relationship context: 11 messages analyzed where **control** appears in 6, **psychological pressure** in 7, **gaslighting** in 6

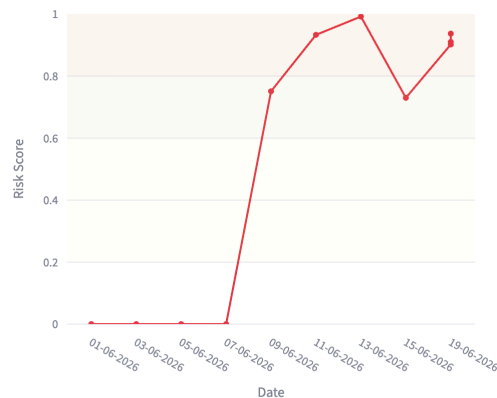
→ Context risk: Medium

SafeGuard AI tracks both message-level and context-level risk because a single message rarely tells the full story.

Risk Score Timeline — Relationship Concerns Case

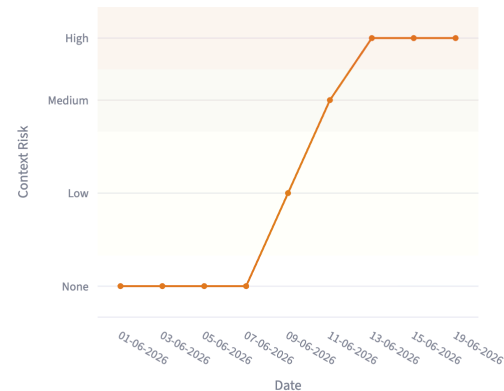
Risk Score Over Time

Individual message risk score based on detected linguistic patterns.



Context Risk Over Time

Risk level reassessed considering the history of previous messages in the case.



Contextual Analysis - Beyond the Single Message

How it works

Before each analysis, the system retrieves:

- Last 3 analyzed messages with risk levels
- Cumulative relationship summary; a GPT-generated behavioral profile updated after every analysis

If toxic patterns are found in recent history:

- Gate threshold lowered (0.92 → 0.75)
- System becomes more sensitive to subtle patterns
- GPT receives full relationship context

What it produces

context_risk_level: separate from **message risk_level**

Example:

"Who were you with tonight?"

→ Message risk: None (isolated)

→ Context risk: Medium (after history of control)

"You are nothing without me."

→ Message risk: Medium (bart classifier)

→ Context risk: High (pattern established)

A single message rarely tells the full story

Responsible AI Design

✓ Pattern analysis only

The system analyzes linguistic patterns, never people. No individual is ever labeled.

✓ PII protection

Microsoft Presidio removes names, phone numbers, emails, IBANs before any external AI call.

✓ Hedged language

GPT is constrained to use "may suggest", "is often associated with", never definitive statements.

✓ Context-aware analysis

Every analysis considers the full relationship history, providing a context risk level alongside the message-level assessment.

✓ RAG-grounded

Every explanation is anchored to retrieved educational documents. No hallucination possible.

✗ No diagnosis

The system never assigns a mental health label, legal judgment, or personal accusation.

Known Limitations & Next Steps

Current Limitations

- Zero-shot classifier may produce false positives on ambiguous language
- Gate model trained on generic data, fine-tuned domain dataset in progress
- Contextual analysis may amplify risk on neutral messages within a toxic history
- English language only
- Sarcasm, emoji, and social media slang not fully supported

Future Improvements

- Domain-specific fine-tuned multi-label classifier
- Context-aware classification using conversation history
- Proprietary training dataset with relational examples
- Multi-language support
- Screenshot OCR and document upload
- Professional portal for counselors
- Deployment on Render + Supabase

What This Project Demonstrates

Generative AI

- Two distinct AI models
- GPT-4o-mini + HuggingFace
- Prompt engineering with ethical policy

RAG Pipeline

- pgvector embeddings
- LangChain retrieval
- Constrained generation

Fine-tuning

- DistilBERT fine-tuned
- civil_comments + daily_dialog
- Custom domain examples

Full-Stack Dev

- FastAPI REST API
- PostgreSQL + pgvector
- Streamlit frontend

Responsible AI

- PII anonymisation
- Ethical system prompt
- No diagnosis principle

NLP & Analytics

- Multi-label classification
- Behavioral timeline
- Contextual relationship analysis
- Pattern trend detection



SafeGuard AI

Thank you

github.com/MapiAI/SafeGuard-AI